
9.8 KEY WORDS

Generalised Least Squares : GLS is basically application of OLS to the transformed model which satisfies the classical assumption. In this method, the variables are transformed by incorporating the autocorrelation parameter ' ρ '. The GLS procedure produces efficient estimates since all available information (about autocorrelation or heteroscedasticity) is used. In the presence of autocorrelation, the GLS estimator is BLUE.

Serial correlation or Autocorrelation : In time series data, the correlation between error terms is known as autocorrelation or serial correlation.

White Noise : An error term ' ε_t ' which satisfies the following properties:

$$E(\varepsilon_t) = 0$$

$$Var(\varepsilon_t) = \sigma_\varepsilon^2$$

$$Cov(\varepsilon_t, \varepsilon_{t+s}) = 0; s \neq 0$$

9.9 SOME USEFUL BOOKS/REFERENCES

Brooks, Chris (2014), '*Introductory Econometrics for Finance*,' Third Edition, Cambridge University Press

Gujarati, Damodar N. and Sangeetha (2007), '*Basic Econometrics*,' Fourth Edition, Tata McGraw Publishing Company Limited, New Delhi, Edition 2007

Maddala, G. S. and K. Lahiri (2009), '*Introduction to Econometrics*,' Fourth Edition, John Wiley and Sons

Wooldridge, Jeffrey (2022), '*Introductory Econometrics*,' Seventh Edition, Cengage Learning

9.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) False. The consequences of autocorrelation are given in Section 9.3.
- 2) True. Refer to Section 9.3.
- 3) You should represent u_t in terms of u_{t-2} and find the covariance between the two.

Check Your Progress 2

- 1) The procedure of DW test is given in Section 9.4.
- 2) The limitations of DW test are given in Section 9.4.

3) Yes, there is positive autocorrelation. Go through Section 9.4 for details.

Autocorrelation

Check Your Progress 3

- 1) Refer to Section 9.5 and Section 9.3.
- 2) In models having lagged explained variables, the DW test cannot be used.



UNIT 10 MULTICOLLINEARITY*

Structure

- 10.0 Objective
- 10.1 Introduction
- 10.2 Concept of Multicollinearity
- 10.3 Consequences of Multicollinearity
- 10.4 Detection of Multicollinearity
 - 10.4.1 Various Tests to Detect Multicollinearity
 - 10.4.2 Problems in Measurement of Multicollinearity
- 10.5 Remedies for Multicollinearity
- 10.6 Let Us Sum Up
- 10.7 Key Words
- 10.8 Some Useful Books/ References
- 10.9 Answers/ Hints to Check Your Progress Exercises

10.0 OBJECTIVES

After reading this unit, you should be in a position to

- explain the problem of multicollinearity in regression analysis;
- identify how the problem of multicollinearity affects the results of regression analysis;
- explain various methods used to detect multicollinearity in a regression model; and
- suggest remedial measures in case multicollinearity is present in a dataset.

10.1 INTRODUCTION

Multicollinearity is basically the presence of high correlation between explanatory variables in a multiple regression model. An assumption of the OLS method, as you may recall from Unit 3, is that the explanatory variables are not correlated with one another (in technical terms, they are orthogonal). A non-zero correlation between explanatory variables is typically observed. Low correlation between explanatory variables may not be a problem, but if the correlation *between explanatory variables is found to be very high, it causes serious*

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problems in parameter estimation by inflating the standard errors and making t-ratios statistically insignificant.

10.2 CONCEPT OF MULTICOLLINEARITY

As pointed out above, multicollinearity is high intercorrelation between the explanatory variables. Two types of multicollinearity need to be discussed: *perfect multicollinearity* and *near multicollinearity*.

Perfect Multicollinearity: In this case two or more explanatory variables bear an exact linear relationship. In the presence of perfect multicollinearity, estimating all the coefficients of the model is not possible. For example, two variables ' x_2 ' and ' x_3 ' used in a regression model are such that:

$$x_2 = 2 x_3 \quad \dots (10.1)$$

Here, x_2 and x_3 are perfectly related to each other. In this case if we regress x_2 on x_3 we find that the coefficient of determination r_{23}^2 is equal to 1. A scatter plot of x_2 against x_3 will show all points to remain on a straight line. To cite another example, suppose in a regression model (in deviation form)

$$y = \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u \quad \dots (10.2)$$

the independent variables are related to each other by the equation $x_3 = 2x_1 + 3x_2$

If we substitute $x_3 = 2x_1 + 3x_2$ in equation (10.2), we obtain

$$y = \beta_1 x_1 + \beta_2 x_2 + \beta_3 (2x_1 + 3x_2) + u, \quad \text{or} \\ y = (\beta_1 + 2\beta_3) x_1 + (\beta_2 + 3\beta_3) x_2 + u \quad \dots (10.3)$$

Note that equation (10.3) is regression model where y is regressed on x_1 and x_2 with parameters $(\beta_1 + 2\beta_3)$ and $(\beta_2 + 3\beta_3)$. Thus it is possible to estimate the linear combinations of the parameters $\beta_1 + 2\beta_3$ and $\beta_2 + 3\beta_3$, but it is not possible to estimate β_1, β_2 and β_3 separately. That is, when multicollinearity is present in a regression model, disentangling the separate effect of each explanatory variable on the explained or dependent variable is difficult. In a m -variable regression model, there is an exact linear relationship between the variables if,

$$\varphi_1 x_1 + \varphi_2 x_2 + \dots + \varphi_m x_m \quad \dots (10.4)$$

The parameters $\varphi_1, \varphi_2 \dots \varphi_m$ are constants, not all zero simultaneously.

Near Multicollinearity: In this case, correlation between the explanatory variables is high but not perfect (r_{23}^2 is not exactly, but nearly equal to 1). In this case, the scatter plot of x_2 against x_3 will show all points lying close to the straight line, but not exactly on it. In a m -variable regression model, multicollinearity is less than perfect if,

$$\varphi_1 x_1 + \varphi_2 x_2 + \dots + \varphi_m x_m + v \quad \dots (10.5)$$

where v is a stochastic error term.

You should note that in multicollinearity problem we consider only linear relationships between the explanatory variables. It does not apply to non-linear relationships between variables.

10.3 CONSEQUENCES OF MULTICOLLINEARITY

Even with near multicollinearity, the estimators obtained by OLS are:

- Unbiased
- Consistent
- Best Linear Unbiased Estimate (BLUE)

Some of the consequences of multicollinearity are discussed below.

1. Regression coefficients are indeterminate or not precisely estimable

With perfect multicollinearity, the parameters of the explanatory variables cannot be determined. For example, let us consider the regression model

$$y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{u}_i \quad \dots(10.6)$$

Here, y , x_1 and x_2 are taken in deviation form from the respective means, $\hat{\beta}_1$ and $\hat{\beta}_2$ are estimated regression coefficients and ' i ' range from 1 to n . Suppose there is perfect multicollinearity problem such that $x_{2i} = \tau x_{1i}$ (where τ is a constant). Thus the above regression model can be written as

$$y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 (\tau x_{1i}) + \hat{u}_i, \quad \text{or}$$

$$y_i = (\hat{\beta}_1 + \hat{\beta}_2 \tau) x_{1i} + \hat{u}_i \quad \dots (10.7)$$

$$\text{Let } \hat{\beta}_1 + \hat{\beta}_2 \tau = \alpha \quad \dots (10.8)$$

In the above formulation it is not possible to get the estimators for $\hat{\beta}_1$ and $\hat{\beta}_2$ separately. However, $\alpha (= \hat{\beta}_1 + \hat{\beta}_2 \tau)$ can be estimated which is a linear function of the two regression coefficients, $\hat{\beta}_1$ and $\hat{\beta}_2$. It is worth mentioning that we can obtain a unique value for α but there will be infinite solutions for $\hat{\beta}_1$ and $\hat{\beta}_2$, given the values of α and τ . Thus obtaining a unique set of values for $\hat{\beta}_1$ and $\hat{\beta}_2$ is not possible. With less than perfect multicollinearity, the regression coefficients can be estimated but not precisely, because their standard errors turn out to be very high (see discussion below).

2. Standard errors of the coefficients are very high or infinite

The variance and covariance of the estimators are very large in the presence of multicollinearity. With perfect multicollinearity, the standard errors of the estimators reach a value of infinity. With less than perfect multicollinearity, the standard errors of the regression coefficients are very large. In the regression model $y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{u}_i$, the variances of the estimated regression coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ are given by

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum x_{1i}^2 (1 - r_{12}^2)}, \text{ and}$$

$$Var(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_{2i}^2(1-r_{12}^2)} \dots(10.9)$$

The covariance of $\hat{\beta}_1$ and $\hat{\beta}_2$ is given by

$$Cov(\hat{\beta}_1, \hat{\beta}_2) = \frac{-r_{12}^2 \sigma^2}{(1-r_{12}^2)\sqrt{\sum x_{1i}^2 \sum x_{2i}^2}} \dots (10.10)$$

Here, σ^2 is the variance of the error term u_i and r_{12} is the correlation coefficient between explanatory variables x_1 and x_2 . From (10.9) and (10.10), we find that as r_{12} tends to unity (implying perfect multicollinearity), the variance and covariance of the estimators tend to infinity.

3. Confidence intervals are wider

Recall from Unit 3 that confidence interval depends on the size of the standard error. If the standard error for an estimator is high, its confidence interval is wide and the null hypothesis is readily accepted. There is a greater likelihood of Type II error (the hypothesis is false but the test statistic does not reject it).

4. R² is high but few t-ratios are significant

In the presence of multicollinearity, isolating the explanatory power of each independent variable becomes difficult. The R^2 is found to be high, but the individual variable's regression coefficients are found to exhibit low levels of statistical significance or they may be statistically insignificant.

You may recall that $t = \frac{\hat{\beta}}{se(\hat{\beta})}$. If there is an increase in the standard error of $\hat{\beta}$, there is a decrease in its t-ratio. In the presence of multicollinearity, as individual regression coefficients exhibit high standard errors, the t-ratios become smaller and statistically not significant. The null hypothesis of all slope coefficients being jointly zero is rejected on the basis of the F-test (reflected in the high R-squared value), but individual regression coefficients are declared insignificant on the basis of the t-test.

5. Regression results become very sensitive to data or specification changes

In the presence of multicollinearity, addition or removal of any independent variable causes large changes in the values of regression coefficients and their standard errors.

Check Your Progress 1

- 1) “In multiple regression, a high correlation in the sample among the regressors implies that the least square estimators of the coefficients are biased”. True or False? Justify your answer.

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- 2) You are given a regression model $Y = \alpha + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 (X_{2i} + X_{3i}) + \beta_5 X_{2i}^2 + \beta_6 X_{3i}^2 + u_i$. Which of the parameters in this model can be estimated by OLS?

.....

10.4 DETECTION OF MULTICOLLINEARITY

There are a few rules of thumb for detecting multicollinearity. Here, the idea is not to see if multicollinearity is present or absent, but to measure its degree or strength in the sample at hand.

10.4.1 Various Methods to Detect Multicollinearity

There are several methods to detect the presence of multicollinearity.

1. Correlation Matrix

Let us assume that a regression model has three explanatory variables, viz. ' x_1 ', ' x_2 ' and ' x_3 '. A matrix of pair-wise correlations among the above variables is given as follows:

correlation	x_1	x_2	x_3
x_1	-	0.1	0.85
x_2	0.1	-	0.3
x_3	0.85	0.3	-

Here, a high correlation between ' x_1 ' and ' x_3 ' shows the presence of multicollinearity. However, such a simple method of detection of multicollinearity will not prove useful if three or more variables are collinear. For example, suppose, $x_1 + x_2 = x_3$. In this case, the coefficient of determination (R^2) obtained from regressing x_3 on x_1 and x_2 equals unity. However, the pair-wise correlations between x_3 and x_1 , between x_1 and x_2 and between x_1 and x_2 may not be high. Thus, pair-wise correlations are not a useful indicator of multicollinearity when three or more variables are collinear.

2. High Standard Errors of the Regression Coefficients

Due to high inter-correlation among independent variables, the estimated regression coefficients have high standard errors. But this is not necessarily true as demonstrated below. Even low inter-correlation among independent variables can cause high standard errors. In the regression model $y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{u}_i$ given above, the variance of the estimated regression coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ are given by the following:

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum x_{1i}^2(1-r_{12}^2)}$$

$$Var(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_{2i}^2(1-r_{12}^2)}$$

We can see that the variance of $\hat{\beta}_1$ will be high if i) σ^2 is high, ii) $\sum x_{1i}^2$ is low, and iii) r_{12}^2 is high. Thus, just knowing that r_{12}^2 is high, does not tell us whether multicollinearity exists or not. In fact, standard errors can be high if σ^2 is high and $\sum x_{1i}^2$ is low. Hence, the proposition that high pair-wise correlation between explanatory variables necessarily cause high standard errors is not true. Extending the above formula to a k -variable regression model, the variance of the j^{th} regression coefficient is given by:

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{\sum x_j^2(1-R_j^2)} \quad \dots (10.11)$$

Here, $\sum x_j^2 = \sum (X_j - \bar{X})^2$

$\hat{\beta}_j$ is the estimator of β_j the regression coefficient of explanatory variable X_j . R_j^2 is the R^2 obtained from the regression of X_j on the remaining explanatory variables. Alternatively, R_j^2 is the squared multiple correlation coefficient between X_j and other regressors. From (10.11) we can see that “high R_j^2 is neither necessary nor sufficient condition of getting high standard errors. Thus multicollinearity by itself need not cause high standard errors” (Maddala, 2001).

3. Variance Inflation Factor (VIF)

Another indicator of multicollinearity is the VIF. In the regression model $y_i = \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \hat{u}_i$ given above, the VIF is given by:

$$VIF = \frac{1}{(1-r_{12}^2)} \quad \dots (10.12)$$

In the case of a multiple regression model, the VIF is given by:

$$VIF_j = \frac{1}{(1-R_j^2)} \quad \dots (10.13)$$

You can observe from (10.12) that VIF tends to ∞ as r_{12}^2 tends to 1. With zero correlation between the independent variables, the VIF is unity. That is, higher intercorrelation between the explanatory variables scales up the VIF and the variances and standard errors of regression coefficients. As seen from the formulas below, $Var(\hat{\beta}_1)$ and $Var(\hat{\beta}_2)$ increases as VIF increases.

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum x_{1i}^2} VIF \quad \dots (10.14)$$

$$Var(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_{2i}^2} VIF \quad \dots (10.15)$$

However, as already discussed, high VIF need not necessarily cause high standard error for $\hat{\beta}_1$. If σ^2 is low and $\sum x_{1i}^2$ is high (or, $\sum x_{2i}^2$ is high) we obtain high variance and standard error. Whether multicollinearity is posing a problem or not can be best understood by looking at the standard errors and t-ratios. For $\hat{\beta}_2$, the high intercorrelation among the independent variables may not cause high standard errors. Thus, VIF can be a misleading indicator of the extent

to which multicollinearity is posing a problem. Moreover, VIF has to be computed separately for each explanatory variable. In this sense, VIF is rather useful in identifying the variables that are highly correlated with other explanatory variables and hence can be dropped.

4. Tolerance

The inverse of VIF is known as Tolerance (TOL) and it is another indicator of collinearity between variables.

$$TOL_j = \frac{1}{VIF_j} = (1 - R_j^2) \quad \dots (10.16)$$

TOL_j ranges from 0 (perfect collinearity) to 1 (no collinearity).

5. Klien's Rule of Thumb

Klein argued that multicollinearity is a problem only if R_j^2 exceeds the overall R^2 . This condition can be stated as:

$$R_j^2 > R^2 \quad \dots (10.17)$$

However, this rule has limitations as it is possible to obtain significant regression coefficients even if $R_j^2 > R^2$.

6. Condition Number (CN)

Condition number is an overall measure of multicollinearity. Remember that if the explanatory variables are exactly linearly dependent, the matrix of explanatory variables $X'X$ is singular. $X'X$ is close to being singular if the variables are almost linearly dependent; even if not exactly linearly dependent. The 'condition number' measures multicollinearity from this perspective. In a k -variable regression model, the matrix $X'X$ of explanatory variables is used to find the maximum and minimum eigen value. Eigen values or characteristic roots of any $n \times n$ matrix A are obtained by solving for γ from the characteristic equation $|A - \gamma I| = 0$. This is an n^{th} degree equation in γ having 'n' characteristic roots. The word 'eigen' is borrowed from German, which means 'proper'.

$$\text{Condition Number} = \sqrt{\frac{\text{maximum eigen value}}{\text{minimum eigen value}}} \quad \dots (10.18)$$

Condition number can be calculated by taking the maximum eigen value and any other eigen value. In relation to the maximum eigen value, if the other eigen value is low, it shows the presence of linearly-dependent explanatory variables in the data or near-linear dependencies. The condition number (CN) is equal to 1 if there is no multicollinearity. The condition number increases with increase in collinearity. There are some informal rules of thumb with respect to CN. If the CN is 15, multicollinearity is moderate. If the CN exceeds 30, multicollinearity is severe. By suitably transforming the variables, the CN can be made close to 1.

7. Theil's Measure of Multicollinearity

VIF and CN do not look at the intercorrelation between dependent and independent variables. They only examine the intercorrelation between

independent variables. Theil's Measure of Multicollinearity (m') examines the correlation between the explained variable and the explanatory variables. It is indicated by:

$$m' = R^2 - \sum_{p=1}^k (R^2 - R_{-p}^2) \quad \dots (10.19)$$

Here, R^2 is the squared multiple correlation obtained by regression of 'y' on 'k' explanatory variables. R_{-p}^2 is the squared multiple correlation obtained by regression of 'y' on 'k' explanatory variables omitting x_p . $(R^2 - R_{-p}^2)$ is the incremental contribution of the p^{th} variable to R^2 . The incremental contribution of 'k' explanatory variables ($x_1, x_2, \dots, x_k$) add up to R^2 if these variables are mutually uncorrelated. In that case, $m' = 0$. But with multicollinearity, ' m' ' can be highly positive or even negative, making it a not so useful indicator of the problem of multicollinearity.

10.4.2 Problems in Measurement of Multicollinearity

In order to understand the problems in measurement of multicollinearity, consider the following example where the independent variables can be defined in various ways. Defining C (per capita real consumption), Y (per capita real income), Y_P (per capita real permanent income) and Y_T (per capita real transitory income) we have the relationship:

$$Y = Y_P + Y_T \quad \dots (10.20)$$

Let us assume that Y_T and Y_P are not correlated, i.e., $E(y_T, y_P) = 0$

The consumption function is given by any of the following three ways:

$$C = \alpha Y + \beta Y_P + u \quad \dots (10.21)$$

By substituting (10.20) in (10.21) we obtain

$$C = \alpha Y_T + (\alpha + \beta) Y_P + u \quad \dots (10.22)$$

The above can be expressed as

$$C = (\alpha + \beta) Y - \beta Y_T + u \quad \dots (10.23)$$

The above three ways (equations 10.21, 10.22 and 10.23) in which the consumption function is expressed are essentially the same. But multicollinearity may be reported to be high in the first expression and there may be no multicollinearity in the second expression since it is given that Y_T and Y_P are not correlated. What matters is how precisely we can estimate α , β or $(\alpha + \beta)$. In fact, in some cases, the linear combination $(\alpha + \beta)$ is more precisely estimable than α or β . Such linear combinations often make economic sense based on theory.

Thus, when examining whether multicollinearity is a serious issue, we should first examine the pair-wise correlation between various explanatory variables. High correlation may be suggestive of a problem although not necessarily. If the